Muhammad Mahasin Irfan

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Research Interests

System software; performance measurement; memory systems; heterogeneous machine learning execution.

Education

Lahore University of Management Sciences (LUMS) Lahore, Pakistan

B.S. in Computer Science; CGPA: 3.85/4.00 *September 2023 – August 2027*

- Dean's Honor List in every completed academic year (2023–24, 2024–25, 2025–26).

Workshop Paper

M. M. Irfan, M. S. Hassan, and F. Zaffar. "Decomposing Error in Continual World Models." **NeurIPS 2026 Workshop on Continual World Models**, Idea Track. *Accepted*. PDF.

Selected Research Experience

Heterogeneous GPU/PIM Workload Splitting November 2025 – Present

Undergraduate Researcher, LUMS *Lahore, Pakistan*

Advisors/Collaborators: Dr. Fareed Zaffar; Humza Ikram (CMU); Dr. Hashim Sharif (AMD Research)

- Built repeatable GPU/PIM measurement pipelines across machines, including scripts and Docker environments; evaluated configurations on four CNNs and four LLMs.
- Extended the Newton PIM model in Ramulator using C++ to test additional bank groups and wider rows with corresponding t_{FAW} changes; measured 1.3× and 1.5× simulated throughput improvements over the original configuration.
- Developed a prediction guided split search that evaluates up to 7–8× fewer candidates while selecting configurations within 1% of exhaustive oracle end-to-end latency in the best evaluated cases. Tested limitations on nonmonotonic TensorRT curves.

Evaluation of Continual World Model Updates July 2026 – Present

First Author Researcher, LUMS *Lahore, Pakistan*

Advisors/Collaborators: Dr. Fareed Zaffar

- Derived an exact decomposition of post-update prediction error into inherited error and update mismatch, and wrote the first draft of the accepted workshop paper.
- Designed controlled experiments across prediction horizons; found opposite endpoint/mismatch rankings in 544/544 retained one-step linear trials and reproduced the phenomenon across 500 learned-model runs.

Diffusion Model Quantization Framework June 2025 – August 2025

AI Optimization Intern, 10x Engineers *Lahore, Pakistan*

Advisors/Collaborators: Dr. Saad bin Nasir; Dr. Khurram Usman Mazher

- Extended QuantX for simulated post-training quantization of diffusion pipelines with component aware module traversal, configurable Linear and Conv2d wrappers, and activation collection across denoising steps.
- Debugged misleading FID and latency comparisons by changing evaluation datasets and sample counts, matching prompts and latents, and isolating the effect of `torch.compile`.
- Ablations showed that excluding convolution quantization reduced W4 LPIPS from 0.5713 to 0.3380; subsequent smoothing improved it to 0.3357. Project notes are linked from the website.

Teaching Experience

Teaching Assistant, Computer Systems and Assembly Language (CS225) Fall 2026

LUMS; instructors: Dr. Zartash Afzal Uzmi and Dr. Jahangir *~350 students*

- Develop quizzes, assignments, and a new systems laboratory; grade examinations and conduct tutorials.

Teaching Assistant, Foundations of Data Science Spring 2026

LUMS; instructor: Dr. Mobin Javed *~200 students*

- Designed Python and data visualization assignments, automated tests, and Jupyter/Otter-Grader workflows.

Teaching Assistant, Introduction to Programming (CS100) Spring 2026

LUMS; instructor: Dr. Fareed Zaffar *~180 students*

- Created assignments, held programming labs and a tutorial, and provided office hours.

Academic Service

Selected Reviewer, NeurIPS 2026 Workshop on Continual World Models 2026

Selected by the workshop Technical Program Committee

- Reviewed one anonymous submission and sent a written review and recommendation to the program committee.

Technical and Research Skills

Python, C++, C, Linux, Docker, Git, PyTorch, Ramulator, TVM, TensorRT, performance benchmarking, architecture simulation, model quantization.

Selected Honors

Dean's Honor List (all completed years) | National Mathematics Talent Contest, Pakistan top 20 | SEAMO Gold Medal, third in Pakistan | Purple Comet! Math Meet, first-place team in Pakistan